to grasp what a smeared electron actually is. As an alternative, in 1926 German physicist Max Born sharply refined Schrödinger's interpretation of an electron wave, and it is his interpretation—amplified by Bohr and his colleagues—that is still with us today. Born's suggestion is one of the strangest features of quantum theory, but is supported nonetheless by an enormous amount of experimental data. He asserted that an electron wave must be interpreted from the standpoint of probability. Places where the magnitude (a bit more correctly, the square of magnitude) of the wave is large are places where the electron is more likely to be found; places where the magnitude is small are places where the electron is less likely to be found. An example is illustrated in Figure 4.9.

This is truly a peculiar idea. What business does probability have in the formulation of fundamental physics? We are accustomed to probability showing up in horse races, in coin tosses, and at the roulette table, but in those cases it merely reflects our incomplete knowledge. If we knew precisely the speed of the roulette wheel, the weight and hardness of the white marble, the location and speed of the marble when it drops to the wheel, the exact specifications of the material constituting the cubicles and so on, and if we made use of sufficiently powerful computers to carry out our calculations we would, according to classical physics, be able to predict with certainty where the marble would settle. Gambling casinos rely on your inability to ascertain all of this information and to do the necessary calculations prior to placing your bet. But we see that probability as encountered at the roulette table does not reflect anything particularly fundamental about how the world works. Quantum mechanics, on the contrary, injects the concept of probability into the universe at a far deeper level. According to Born and more than half a century of subsequent experiments, the wave nature of matter implies that matter itself must be described fundamentally in a probabilistic manner. For macroscopic objects like a coffee cup or the roulette wheel, de Broglie's rule shows that the wave-like character is virtually unnoticeable and for most ordinary purposes the associated quantum-mechanical probability can be completely ignored. But at a microscopic level we learn that the best we can ever do is say that an electron has a particular probability of being found at any given location.